

Project P04 - Intelligent Trademark Registration Platform

ID: P04 | version: v2 | asset_type: project | language: en
sector: Public sector | duration: 16 months
stack: Python, NLP, LLM, RAG, Elasticsearch, Azure, Docker, MLflow

Client

Government agency D

Description

Guided-filing chatbot and a crawler detecting similar trademarks and patents.

Measurable result

Processing time down 40%

Context

Government agency handling a growing volume of trademark and patent applications with long manual checks.

Challenges

Complex registry rules, multilingual documents and strict hosting constraints.

Solution

RAG chatbot grounded on registry rules plus a crawler using embedding and keyword search to flag similar marks.

Deliverables

- Guided-filing chatbot
- Similarity crawler and index
- Azure reference architecture
- Evaluation report

Team

- C11 - Dr. Amine Khelifi - AI lead
- C19 - Nabil Al-Mansoori - Solution architect